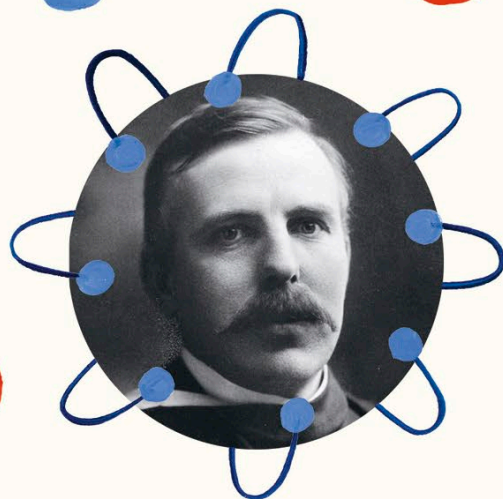




ERNEST RUTHERFORD

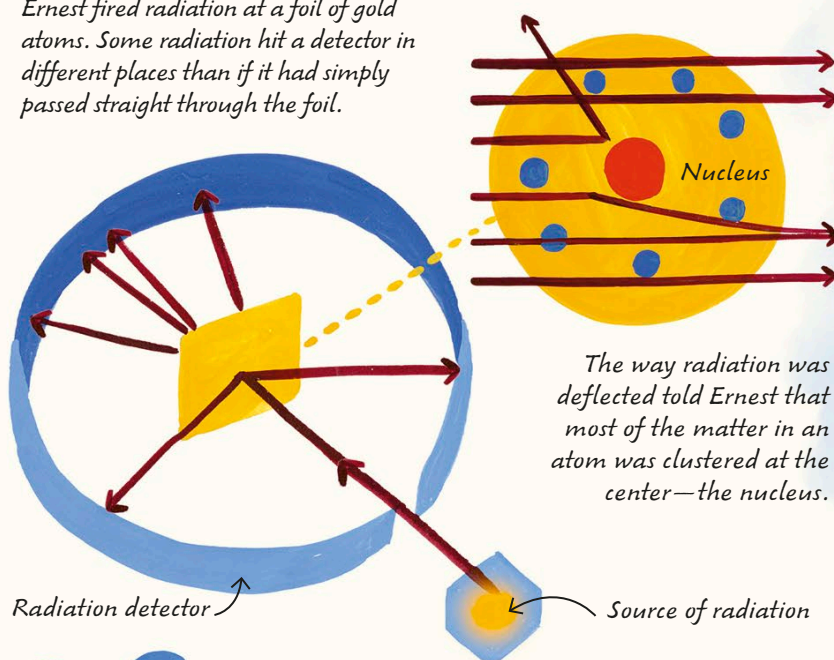
New Zealand-born British physicist • 1871–1937

Some scientists become famous for one world-changing discovery. Ernest Rutherford made three! The most important was that atoms are mostly empty space. This set the stage for all kinds of new findings about atoms and radiation, a type of energy.

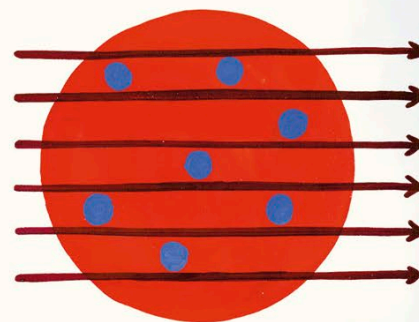
Ernest went on to explain what makes certain things—known as radioactive substances—give off radiation. He also named the proton, one of the three tiny particles that are the building blocks of all atoms.

Having grown up on a farm without easy access to a good scientific education, Ernest worked hard to help other physics students succeed. Many of the students he trained made incredible discoveries themselves. In addition to helping individuals, he also changed the world—by beginning the science of nuclear physics.

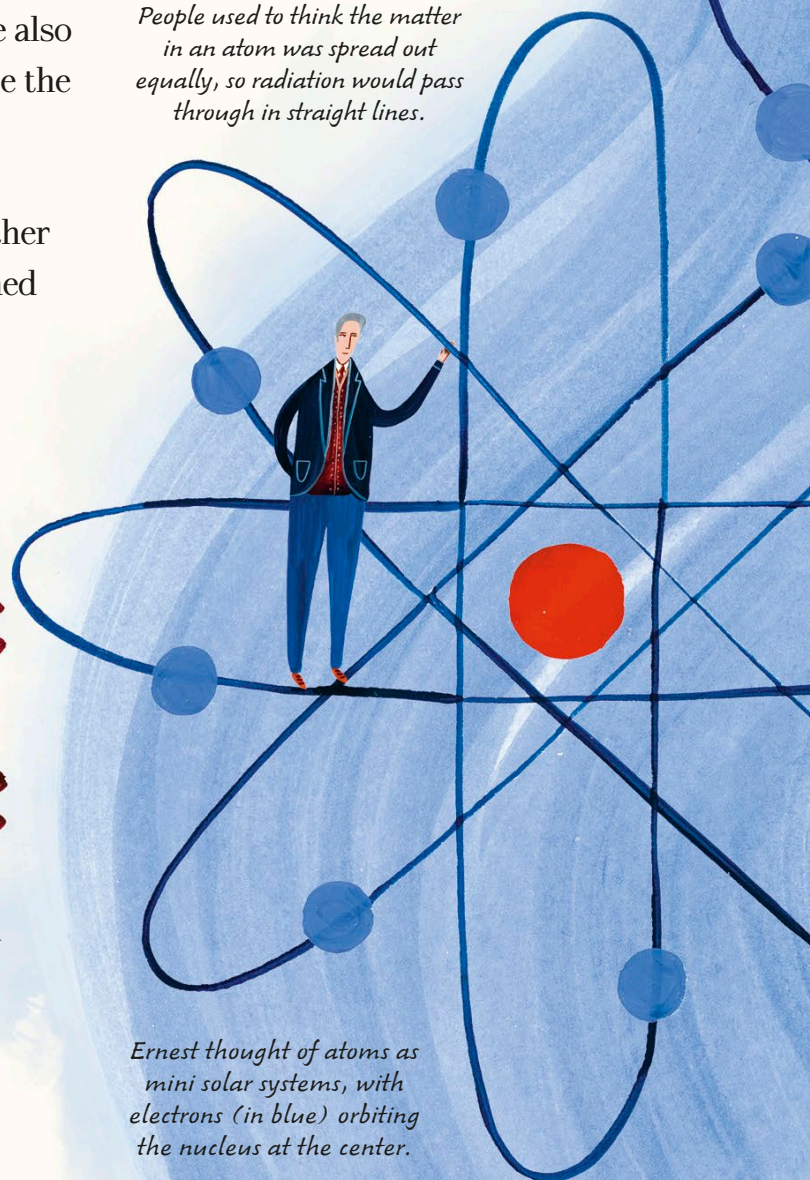
Ernest fired radiation at a foil of gold atoms. Some radiation hit a detector in different places than if it had simply passed straight through the foil.



The way radiation was deflected told Ernest that most of the matter in an atom was clustered at the center—the nucleus.



People used to think the matter in an atom was spread out equally, so radiation would pass through in straight lines.



Ernest thought of atoms as mini solar systems, with electrons (in blue) orbiting the nucleus at the center.